

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 2**Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
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Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A particle undergoes simple harmonic motion described by $x(t) = 0.04 \cos(10\pi t)$, where x is in metres and t is in seconds. What is the magnitude of the maximum acceleration of the particle? [2 marks]

- A. $0.4\pi \text{ m/s}^2$
- B. $0.4\pi^2 \text{ m/s}^2$
- C. $4\pi^2 \text{ m/s}^2$
- D. $40\pi^2 \text{ m/s}^2$

2. A mass on a spring oscillates with amplitude A . At what displacement from equilibrium is the kinetic energy equal to the potential energy? [2 marks]

- A. $A/4$
- B. $A/2$
- C. $A/\sqrt{2}$
- D. $A/\sqrt{3}$

3. A damped oscillator has a quality factor $Q = 50$. Approximately how many complete oscillations does it take for the energy of the oscillator to decrease to half of its initial value? [2 marks]

- A. 5.5
- B. 11
- C. 25
- D. 50

4. A damped harmonic oscillator is driven by an external sinusoidal force at the resonant frequency $\omega = \omega_0$. What is the phase lag δ of the displacement behind the driving force at this frequency? [2 marks]

- A. 0
- B. $\pi/4$
- C. $\pi/2$
- D. π

5. Which of the following expressions is a valid solution to the one-dimensional wave equation $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$? [2 marks]

A. $y = A \sin(kx) \cos(kx)$

B. $y = Ae^{-(x-vt)^2}$

C. $y = Ae^x \cos(vt)$

D. $y = A \sin(kx + \omega t^2)$

6. A wave pulse travels along a light string (impedance $Z_1 = 0.5 \text{ kg/s}$) and reaches a junction with a heavier string (impedance $Z_2 = 1.5 \text{ kg/s}$). What is the amplitude reflection coefficient $r = A_r/A_i$? [2 marks]

A. $+1/2$

B. $-1/2$

C. $+1/3$

D. $-1/3$

7. A sound has an intensity level of $\beta = 85 \text{ dB}$. Taking $I_0 = 10^{-12} \text{ W/m}^2$, what is the intensity of the sound? [2 marks]

A. $3.16 \times 10^{-5} \text{ W/m}^2$

B. $3.16 \times 10^{-4} \text{ W/m}^2$

C. $8.5 \times 10^{-4} \text{ W/m}^2$

D. $8.5 \times 10^{-10} \text{ W/m}^2$

8. A fire engine siren emits sound at frequency $f = 800 \text{ Hz}$. The fire engine moves **away** from a stationary observer at speed $v_s = 40 \text{ m/s}$. Taking the speed of sound in air as $v = 340 \text{ m/s}$, what frequency does the observer hear? [2 marks]

A. 716 Hz

B. 742 Hz

C. 800 Hz

D. 863 Hz

9. An organ pipe is open at one end and closed at the other. Which of the following sets correctly lists the harmonic numbers n that are present in its resonant modes? [2 marks]

A. $n = 1, 2, 3, 4, 5, \dots$

B. $n = 2, 4, 6, 8, 10, \dots$

C. $n = 1, 3, 5, 7, 9, \dots$

D. $n = 0, 1, 2, 3, 4, \dots$

10. For a wave packet in a dispersive medium, which of the following correctly describes the relationship between the group velocity v_g and the phase velocity v_p ? [2 marks]

- A. The group velocity always equals the phase velocity.
- B. The group velocity is always greater than the phase velocity.
- C. The group velocity is the speed at which the envelope of the wave packet propagates, and may differ from the phase velocity.
- D. The group velocity has no physical significance; only the phase velocity determines energy transport.

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. Let $|\psi\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{i}{\sqrt{2}}|2\rangle$ and $|\phi\rangle = \frac{i}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle$, where $\{|1\rangle, |2\rangle\}$ is an orthonormal basis. What is $\langle\phi|\psi\rangle$? [2 marks]

- A. 1
- B. i
- C. 0
- D. $-i$

2. Which of the following states is correctly normalised? (Assume $|a\rangle$ and $|b\rangle$ are orthonormal.) [2 marks]

- A. $|\psi\rangle = \frac{1}{2}|a\rangle + \frac{1}{2}|b\rangle$
- B. $|\psi\rangle = \frac{1}{\sqrt{2}}|a\rangle + \frac{i}{\sqrt{2}}|b\rangle$
- C. $|\psi\rangle = |a\rangle + |b\rangle$
- D. $|\psi\rangle = \frac{1}{\sqrt{3}}|a\rangle + \frac{1}{\sqrt{3}}|b\rangle$

3. Two states are given by $|\alpha\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle$ and $|\beta\rangle = c|1\rangle + \frac{1}{\sqrt{2}}|2\rangle$, where $\{|1\rangle, |2\rangle\}$ is an orthonormal basis and c is a real number. Find the value of c such that $|\alpha\rangle$ and $|\beta\rangle$ are orthogonal. [2 marks]

- A. $c = 1/\sqrt{2}$
- B. $c = -1/\sqrt{2}$
- C. $c = 1/2$
- D. $c = -1$

4. An operator U acting on a quantum state space is unitary. Which of the following properties must U satisfy? [2 marks]

- A. $U^\dagger = U$
- B. $U^\dagger U = I$ (the identity operator)
- C. $U^2 = I$
- D. $U = U^{-1}$

5. A spin-1/2 particle is in the state $|\psi\rangle = \frac{1}{\sqrt{3}}|\uparrow_z\rangle + \sqrt{\frac{2}{3}}|\downarrow_z\rangle$. What is the expectation value $\langle S_z \rangle$? [2 marks]

- A. $+\hbar/6$
- B. $-\hbar/6$
- C. $+\hbar/3$
- D. $-\hbar/3$

6. A spin-1/2 particle is prepared in the state $|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$. What is the probability of measuring $S_z = +\hbar/2$? [2 marks]

- A. 0
- B. 1/4
- C. 1/2
- D. 1

7. A quantum system is prepared in an energy eigenstate $|E_n\rangle$ of the Hamiltonian H at time $t = 0$. How does the probability of measuring an observable A to have eigenvalue a_k change with time? [2 marks]

- A. It oscillates sinusoidally.
- B. It decreases exponentially.
- C. It remains constant.
- D. It increases linearly.

8. Consider two spin-1/2 particles. Which of the following two-particle states is a **product state** (i.e. **not** entangled)? [2 marks]

- A. $\frac{1}{\sqrt{2}}(|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2)$
- B. $\frac{1}{\sqrt{2}}(|\uparrow\rangle_1 |\uparrow\rangle_2 + |\downarrow\rangle_1 |\downarrow\rangle_2)$
- C. $\frac{1}{2}(|\uparrow\rangle_1 + |\downarrow\rangle_1)(|\uparrow\rangle_2 - |\downarrow\rangle_2)$
- D. $\frac{1}{\sqrt{2}}(|\uparrow\rangle_1 |\downarrow\rangle_2 + |\downarrow\rangle_1 |\uparrow\rangle_2)$

9. Two identical bosons occupy a joint quantum state $|\Psi\rangle$. What happens to $|\Psi\rangle$ when the labels of the two particles are exchanged (i.e. the SWAP operator is applied)? [2 marks]

- A. $|\Psi\rangle$ acquires a factor of -1 : $\text{SWAP} |\Psi\rangle = -|\Psi\rangle$
- B. $|\Psi\rangle$ is unchanged: $\text{SWAP} |\Psi\rangle = +|\Psi\rangle$
- C. $|\Psi\rangle$ acquires an arbitrary phase $e^{i\theta}$
- D. $|\Psi\rangle$ becomes zero

10. In beta decay, a neutron transforms into a proton, an electron, and an electron antineutrino: $n \rightarrow p + e^- + \bar{\nu}_e$. Which fundamental force mediates this process? [2 marks]

- A. The gravitational force
- B. The electromagnetic force
- C. The strong nuclear force
- D. The weak nuclear force

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

Rotational Dynamics and Atwood Machine

- (a) A solid cylinder of mass $M = 4$ kg and radius $R = 0.2$ m rolls without slipping down a slope inclined at 30° to the horizontal. Using energy conservation, find the speed of the cylinder after it has descended a vertical height of $h = 3$ m. The moment of inertia of a solid cylinder about its axis is $I = \frac{1}{2}MR^2$. 3
- (b) For the same cylinder on the same slope, find the linear acceleration using Newton's second law. You should write down both the translational and rotational equations of motion and use the rolling constraint $a = R\alpha$. 3
- (c) An Atwood machine consists of two masses $m_1 = 6$ kg and $m_2 = 4$ kg connected by a light, inextensible string passing over a frictionless, massless pulley. Find the acceleration of the system and the tension in the string. 4
- (d) Now suppose the pulley is a solid disk of mass $M_p = 2$ kg and radius $R_p = 0.1$ m. The string does not slip on the pulley. Set up the equations of motion for both masses and the pulley. Hence find the new acceleration of the system. 5

Question 2.

[15 marks]

Ideal Gas, Carnot Engine, and Entropy

- (a) A container holds $n = 3$ moles of a diatomic ideal gas ($\gamma = 7/5$) at temperature $T = 400$ K and pressure $P = 2 \times 10^5$ Pa. Find the volume of the gas and the internal energy U . For a diatomic gas at room temperature, $U = \frac{5}{2}nRT$. 3
- (b) The gas undergoes a reversible adiabatic compression to half its original volume. Find the final temperature and the final pressure. 4
- (c) A Carnot engine operates between a hot reservoir at $T_h = 600$ K and a cold reservoir at $T_c = 300$ K. If the engine absorbs $Q_h = 2000$ J per cycle from the hot reservoir, calculate the work done per cycle, the heat rejected to the cold reservoir, and the efficiency of the engine. 4
- (d) A 2 kg block of copper with specific heat capacity $c = 385$ J/(kg·K) at an initial temperature of 500 K is placed in thermal contact with a large reservoir at temperature $T_R = 300$ K. Calculate the entropy change of the copper block, the entropy change of the reservoir, and the total entropy change of the universe. Comment on the sign of the total entropy change. 4

Question 3.

[15 marks]

Waves on Strings and Sound

- (a) A string of length $L = 1.5$ m and total mass $m = 0.012$ kg is stretched under a tension $T = 75$ N. Calculate the linear mass density μ , the speed v of transverse waves on the string, and the fundamental frequency f_1 for standing waves when the string is fixed at both ends. 3
- (b) Write down the equation $y(x, t)$ for a harmonic wave travelling in the $+x$ direction on this string at the 3rd harmonic frequency, using an amplitude $A = 2$ mm. Give numerical values for the angular frequency ω and wavenumber k . 3
- (c) This string is joined to a heavier string with linear mass density $\mu_2 = 0.032$ kg/m, with the same tension $T = 75$ N in both strings. Calculate the impedance of each string, and the amplitude reflection coefficient r and the amplitude transmission coefficient t for a wave incident from the lighter string. 4
- (d) Calculate the fraction of incident power that is reflected and the fraction that is transmitted at the junction. Verify that they sum to unity. 2
- (e) An organ pipe of length $\ell = 0.85$ m is open at one end and closed at the other. Taking the speed of sound in air as $v_s = 340$ m/s, find the frequencies of the first three resonant modes. 3

Question 4.

[15 marks]

Mach-Zehnder Interferometer and Two-Particle States

In a Mach-Zehnder interferometer, a photon travels along one of two paths labelled $|T\rangle$ (top) and $|B\rangle$ (bottom). A beam splitter acts as follows:

$$|T\rangle \rightarrow \frac{1}{\sqrt{2}}(|T\rangle + i|B\rangle), \quad |B\rangle \rightarrow \frac{1}{\sqrt{2}}(i|T\rangle + |B\rangle).$$

Each mirror adds a phase factor of i to the state reflected from it.

- (a) A photon enters the interferometer in state $|T\rangle$. Apply the first beam splitter, then the mirrors (which multiply each path by i), then the second beam splitter. Find the final state. What is the probability of detection at each output port? 4
- (b) A phase shifter $e^{i\varphi}$ is now inserted in the top path between the mirrors and the second beam splitter. Show that the detection probabilities become $P(T) = \sin^2(\varphi/2)$ and $P(B) = \cos^2(\varphi/2)$. 3

- (c) Consider the two-particle state

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2).$$

Show that this state is entangled (i.e. it cannot be written as a product state $|a\rangle_1 |b\rangle_2$). If S_z is measured on particle 1 and the result $+\hbar/2$ is obtained, what is the state of particle 2? 4

- (d) A spin-1/2 particle in a magnetic field has Hamiltonian

$$H = \frac{E_0}{2}(|\uparrow_z\rangle\langle\uparrow_z| - |\downarrow_z\rangle\langle\downarrow_z|),$$

so that $H|\uparrow_z\rangle = (E_0/2)|\uparrow_z\rangle$ and $H|\downarrow_z\rangle = -(E_0/2)|\downarrow_z\rangle$.

The particle starts at $t = 0$ in the state $|\psi(0)\rangle = |\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$.

- (i) Find $|\psi(t)\rangle$.
- (ii) Calculate the probability of measuring $S_z = +\hbar/2$ as a function of time.
- (iii) Calculate $\langle S_x \rangle(t)$.

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v_1' = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m